Design Assignment

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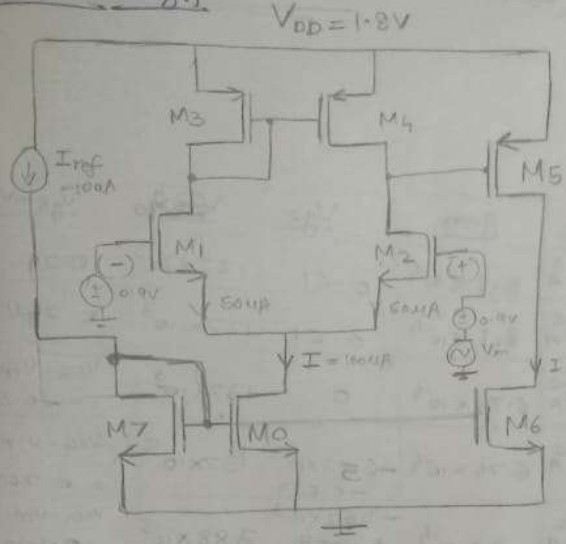
EEE

Initial Design:



Initial Assumptions:

> Giving DC supply of $\left(\frac{V_{DD}}{2}\right)$ as input for amplifier for max swing.

$I_{ref} = 10\mu A$

$\mu_p C_{ox} = 100\mu A$

$\mu_n C_{ox} = 230\mu A$

$V_{ov} = 0.1V$

Initial Calculations:

$$50\mu A = \frac{1}{2} \times 230 \times \left(\frac{W}{L}\right)_{1,2} \times (0.1)^2$$

$$\left(\frac{W}{L}\right)_{1,2} = 43.4782$$

$$50 = \frac{1}{2} \times 100 \left(\frac{W}{L}\right)_{3,4} \times (0.1)^2$$

$$\left(\frac{W}{L}\right)_{3,4} = 100$$

$$100 = \frac{1}{2} \times 230 \times \left(\frac{W}{L}\right)_{0} \times (0.1)^2$$

$$\left(\frac{W}{L}\right)_{0} = 86.956$$

$$\left(\frac{W}{L}\right)_{7} / I_7 = \left(\frac{W}{L}\right)_{0} / I_0 \quad \Rightarrow \left(\frac{W}{L}\right)_{7} = 8.69565$$

(current mirrors)

$$20 = \frac{1}{2} \times k_n \left(\frac{W}{L}\right)_5 \times (0.1)^2 \Rightarrow \left(\frac{W}{L}\right)_5 = 40$$

$$\left(\frac{W}{L}\right)_6 / I_6 = \left(\frac{W}{L}\right)_7 / I_7 \Rightarrow \left(\frac{W}{L}\right)_6 = 17.387$$

		L	W	I_D	gm	V_{gs}	r_o	$V_{gs}-V_{th}$
N	M_1	360 mm	15.65×10^3 nm	4.46×10^{-5} A	8.1×10^{-4}	0.61	127×10^3	0.24 v
N	M_2	360 mm	15.65×10^3 nm	4.46×10^{-5} A	8.1×10^{-4}	0.61	127×10^3	0.24 V
P	M_3	360 mm	36×10^3 mm	4.46×10^{-5} A	6.74×10^{-4}	0	137×10^3	$V_{SG}-V_{TP}$ = −0.39V
P	M_4	360 mm	36×10^3 mm	4.46×10^{-5} A	6.74×10^{-4}	-653×10^{-3} −0.653	137×10^3	$V_{SG}-V_{TP}$ = 0.263V
P	M_5	360 mm	1.45×10^4 mm	2.07×10^{-5} A	3×10^{-4}	-562×10^{-3} −0.562	388×10^3	$V_{SG}-V_{TP}$ = 0.172V
N	M_6	360 mm	6.26×10^3 mm	2.07×10^{-5} A	3.57×10^{-4}	0.542	286×10^3	0.172 v
N	M_7	360 mm	3.1×10^3 mm	1×10^{-5} A	1.74×10^{-4}	0.542	495×10^3	0.172 v
N	M_8	360 mm	31.3×10^3 mm	8.93×10^{-5} A	1.58×10^{-3}	0.542	38×10^3	0.172 v

(r_o is calculated as $1/G_{ds}$) from DC op pt function

→ All Mosfets are in saturation as each satisfies $V_{GS}-V_{Th} < V_{DS}$, $V_{SG}-V_{TP} < V_{SD}$

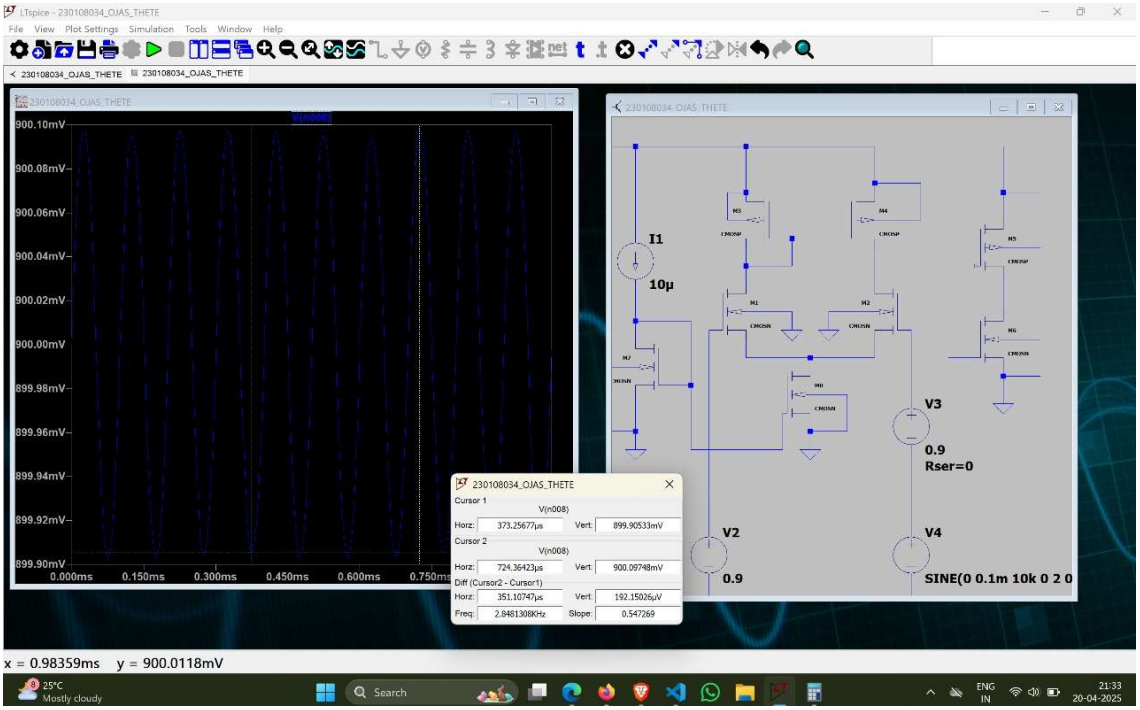
For Single Stage:

When we do DC biasing for the M_1, M_2 Mosfets (0.900 v) we reached a gain of

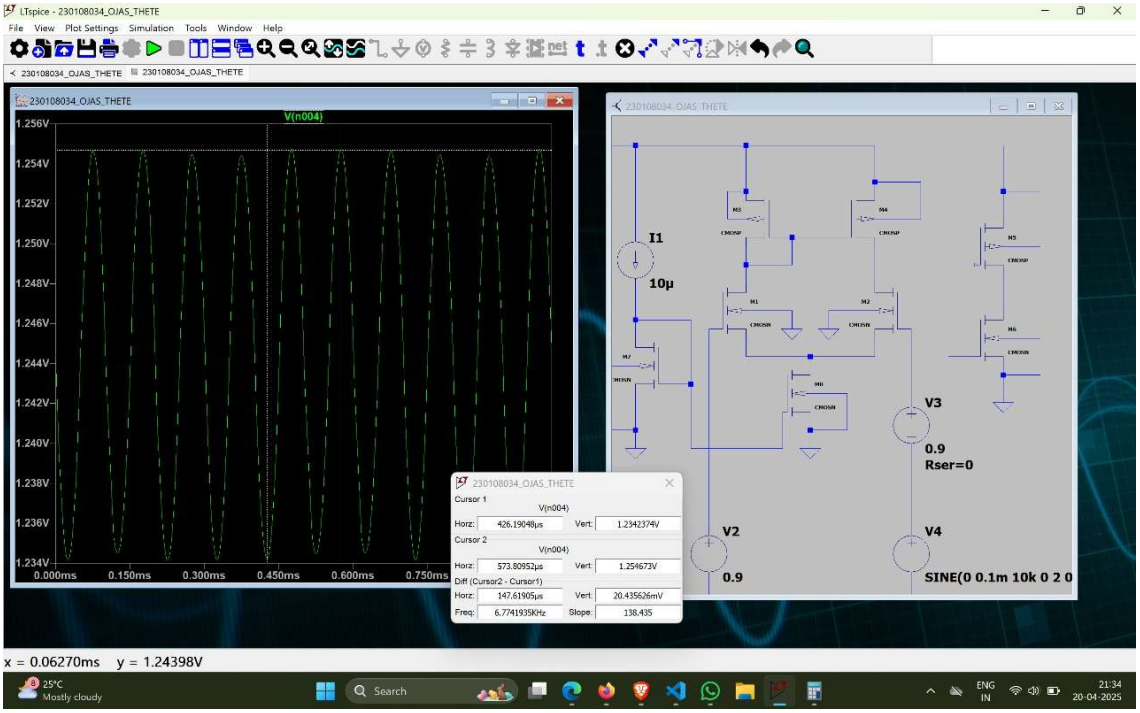
$$\text{Gain} = \frac{V_{out}}{V_{in}} \quad \frac{20.4356\times10^{-3}\ (p-p)}{192.15026\times10^{-6}\ (p-p)} \simeq 106.35$$

when given a sine wave of 0.1 millivolt as input.

Input Peak To Peak Voltage:



First Stage Output:



SPICE Output Log: C:\Users\thete\Documents\LTspice\230108034_OJAS_THETE.log

Semiconductor Device Operating Points:

--- BSIM3 MOSFETS ---

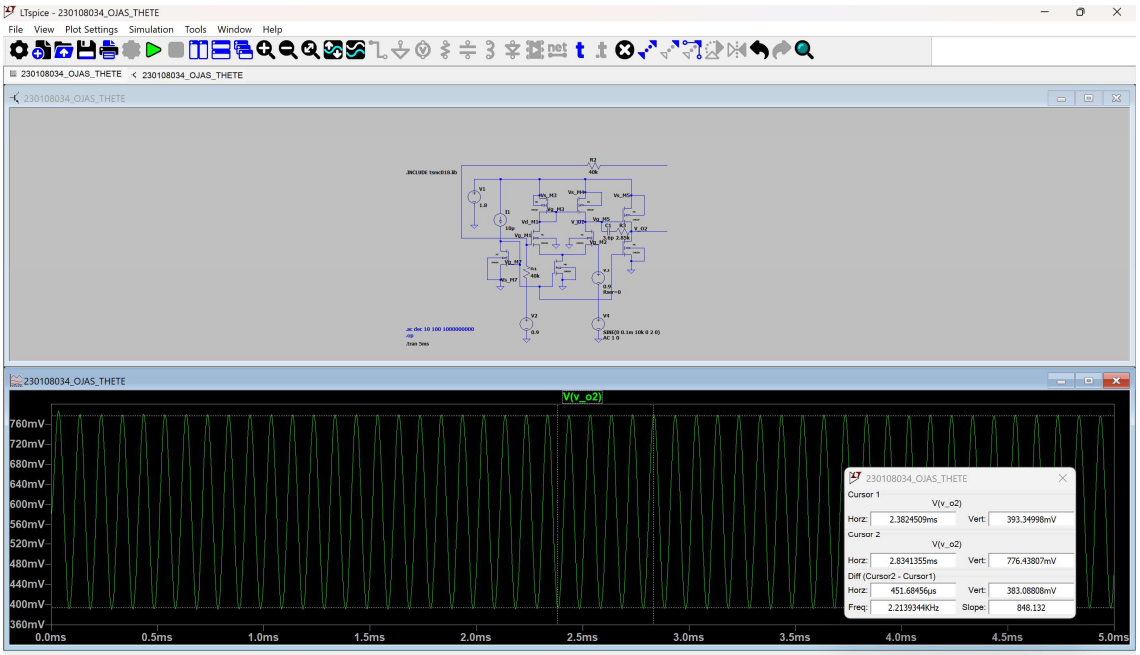
Name:	m3	m4	m5	m1	m2
Model:	cmosp	cmosp	cmosp	cmosn	cmosn
Id:	4.46e-05	4.47e-05	2.07e-05	4.46e-05	4.47e-05
Vgs:	0.00e+00	6.53e-03	3.38e-01	6.10e-01	6.10e-01
Vds:	5.55e-01	5.62e-01	9.00e-01	9.54e-01	9.48e-01
Vbs:	5.55e-01	5.62e-01	9.00e-01	-2.90e-01	-2.90e-01
Vth:	-4.55e-01	-4.55e-01	-4.54e-01	5.43e-01	5.43e-01
Vdsat:	-1.01e-01	-1.01e-01	-1.06e-01	8.03e-02	8.03e-02
Gm:	6.74e-04	6.75e-04	3.00e-04	8.09e-04	8.10e-04
Gds:	7.28e-06	7.24e-06	2.58e-06	7.84e-06	7.85e-06
Gmb	2.03e-04	2.04e-04	9.02e-05	1.99e-04	1.99e-04
Cbd:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cbs:	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.00e+00
Cgsov:	2.29e-14	2.29e-14	9.14e-15	1.29e-14	1.29e-14
Cgdov:	2.29e-14	2.29e-14	9.14e-15	1.29e-14	1.29e-14
Cgbov:	3.01e-19	3.01e-19	3.01e-19	3.26e-19	3.26e-19
dQgdVgb:	1.19e-13	1.19e-13	4.80e-14	5.85e-14	5.85e-14
dQgdVdb:	-2.27e-14	-2.27e-14	-9.08e-15	-1.27e-14	-1.27e-14
dQgdVsb:	-9.28e-14	-9.28e-14	-3.75e-14	-4.25e-14	-4.25e-14
dQddVgb:	-5.13e-14	-5.13e-14	-2.06e-14	-2.54e-14	-2.54e-14
dQddVdb:	2.28e-14	2.28e-14	9.11e-15	1.28e-14	1.28e-14
dQddVsb:	3.74e-14	3.74e-14	1.51e-14	1.59e-14	1.59e-14
dQbdVgb:	-1.67e-14	-1.67e-14	-6.69e-15	-7.69e-15	-7.68e-15
dQbdVdb:	-4.75e-18	-4.16e-18	3.19e-18	9.22e-18	9.21e-18
dQbdVsb:	-4.93e-15	-4.93e-15	-1.97e-15	-2.14e-15	-2.15e-15

SPICE Output Log: C:\Users\thete\Documents\LTspice\230108034_OJAS_THETE.log

dQbdVdb:	-4.75e-18	-4.16e-18	3.19e-18	9.22e-18	9.21e-18
dQbdVsb:	-4.93e-15	-4.93e-15	-1.97e-15	-2.14e-15	-2.15e-15

Name:	m0	m7	m6
Model:	cmosn	cmosn	cmosn
Id:	8.93e-05	1.00e-05	2.07e-05
Vgs:	5.42e-01	5.42e-01	5.42e-01
Vds:	2.90e-01	5.42e-01	9.00e-01
Vbs:	0.00e+00	0.00e+00	0.00e+00
Vth:	4.65e-01	4.62e-01	4.61e-01
Vdsat:	8.06e-02	8.20e-02	8.26e-02
Gm:	1.58e-03	1.74e-04	3.57e-04
Gds:	2.83e-05	2.02e-06	3.50e-06
Gmb	4.35e-04	4.80e-05	9.83e-05
Cbd:	0.00e+00	0.00e+00	0.00e+00
Cbs:	0.00e+00	0.00e+00	0.00e+00
Cgsov:	2.58e-14	2.58e-15	5.15e-15
Cgdov:	2.58e-14	2.58e-15	5.15e-15
Cgbov:	3.26e-19	3.26e-19	3.26e-19
dQgdVgb:	1.19e-13	1.20e-14	2.39e-14
dQgdVdb:	-2.56e-14	-2.55e-15	-5.10e-15
dQgdVsb:	-8.68e-14	-8.75e-15	-1.75e-14
dQddVgb:	-5.15e-14	-5.17e-15	-1.03e-14
dQddVdb:	2.57e-14	2.56e-15	5.12e-15
dQddVsb:	3.37e-14	3.40e-15	6.81e-15
dQbdVgb:	-1.61e-14	-1.62e-15	-3.24e-15
dQbdVdb:	-5.46e-17	1.68e-18	5.35e-18
dQbdVsb:	-6.34e-15	-6.25e-16	-1.25e-15

Second Stage Open Loop Gain:



Open loop Gain $= \frac{383\times10^{-3}}{192\times10^{-6}} = \underline{2k}$

Closed loop Gain $= 1 + \frac{R_f}{R_i} = \underline{2}$

$R_f = R_i = 40K$

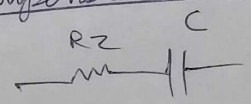
d) <u>Power Consumption:</u>

$P = V(I_{tot}) = (10+100+20)\,\mu A \times 1.8V = \boxed{234\,\mu W}$

(looking at the bottom-most node)

$< \underline{300\,\mu W}$

<u>Miller Compensation & Zero Location:</u>



[C_c is added to separate the poles further apart by Miller compensation. & R_z is added to move the zero towards -ve ∞]

$\Rightarrow g_{m5} - \frac{1}{R_z + \frac{1}{sC}} = 0$

$s = \frac{-g_{m5}}{C(g_{m5}R_z - 1)}$

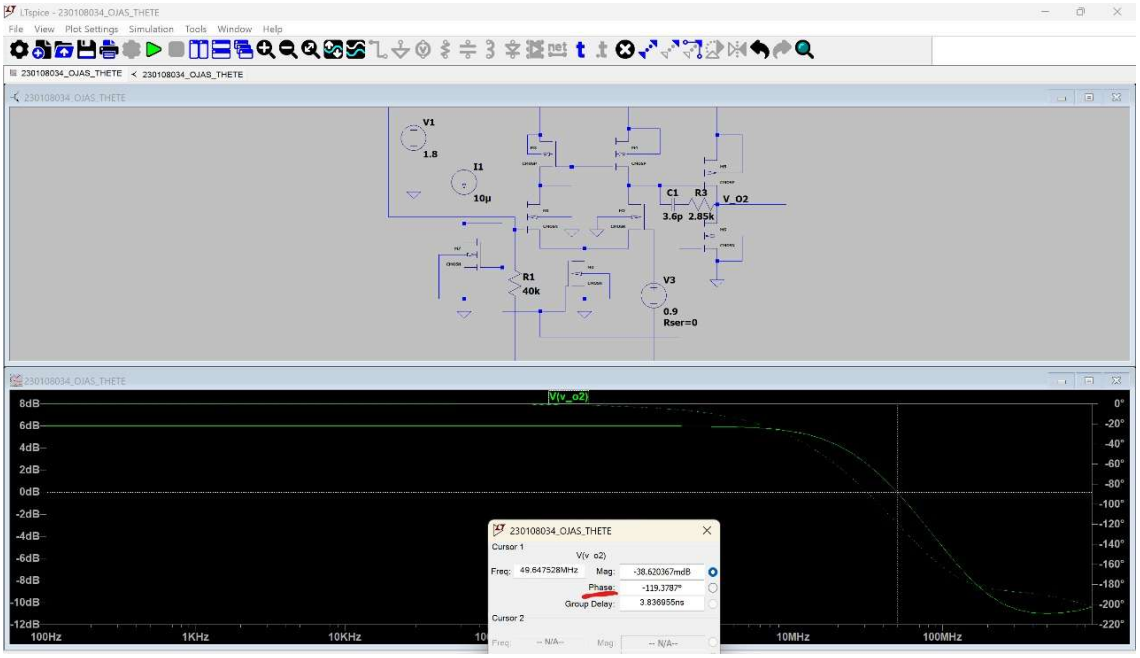
so as $R_z \simeq \frac{1}{g_{m5}}$ zero is shifted

So $\underline{R_z \simeq 3.3K}$

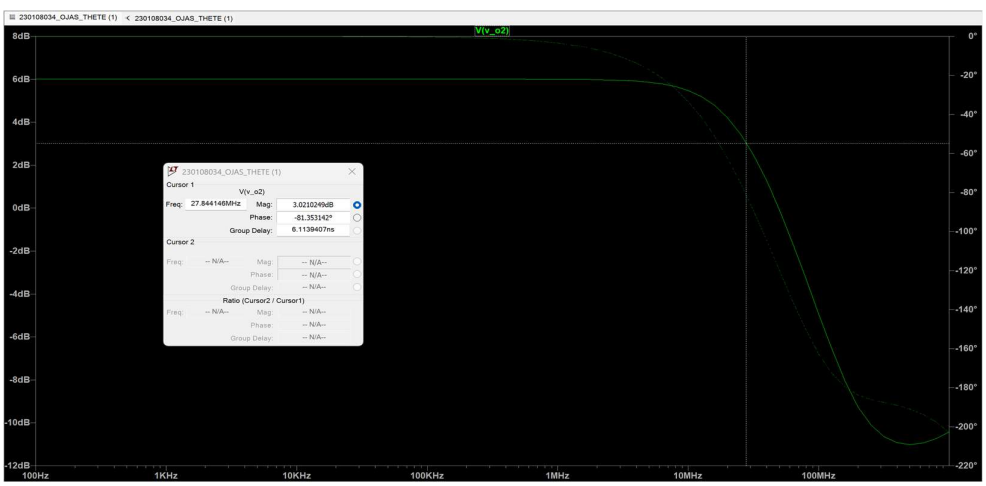
$\Rightarrow$ I have <u>choosen C_c as 3.6pF</u> and varied the R_z (fine-tuning) to get a Phase Margin of $\underline{\simeq 60.6^\circ}$ at 0dB.

R_z value I got $= \underline{2.85K\Omega}$

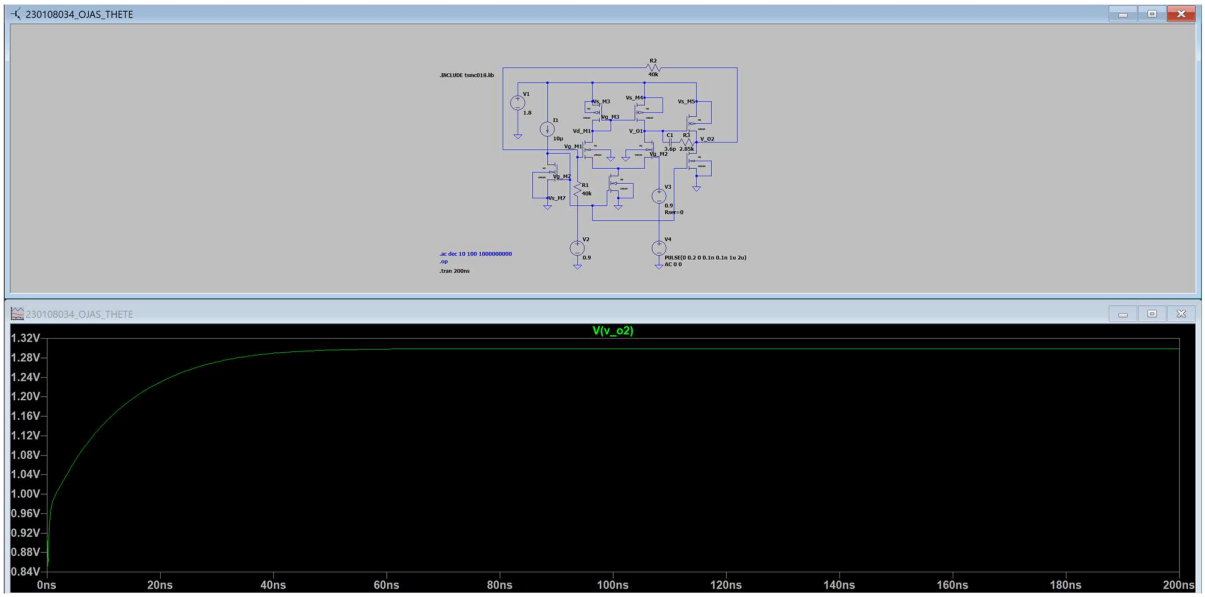
Bode Plot:



Phase Margin = 180+(-119.3787) = 60.6 degrees obtained.



From the above plot 3-dB bandwidth is obtained at 27.88MHz.



Plot of transient response of the closed-loop amplifier with a 0.2V step input (using 0.1ns rise/fall time).

The plot starts at 0.86V and saturates at 1.28V.